rag -- 2026-11-02

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P4] Prevented is Not Protected: What Attack Success Rate Hides about RAG -- It uncovers fundamental flaws in current RAG evaluation practices by separating prevented attacks from introduced failures, offering a concrete benchmark (Aligned RAG Evaluation) that is directly relevant to assessing grounding and faithfulness across RAG systems.

Themes

Several papers focus on improving retrieval quality in multimodal RAG settings. Pix2Evidence fuses screenshot images with transcriptions into a joint encoder and adds a vision-language reranker, yielding up to 4.8-point accuracy gains over PixelRAG on SimpleQA and raising evidence recall by 3-5 points [P1]. AudioHelix tackles long-form audio reasoning with a hierarchical DualGraph that propagates relevance via Personalized PageRank, consistently beating LinearRAG and PlanRAG-Audio across three audio benchmarks, sometimes by more than 2 points in accuracy [P2]. Both works illustrate that richer representations (image-text fusion or graph-based acoustic concepts) can narrow the gap between raw retrieval and downstream generation, though they operate on very different modalities.

A separate line of research examines how the surrounding RAG pipeline influences grounding and faithfulness. The "knowledge compilation" proposal builds a pre-computed navigation tree of claim-evidence pairs to prune context tokens, cutting token usage by 91 % and latency from 140 s to 30 s without changing answer accuracy on FRAMES, highlighting that systematic context structuring can improve efficiency even if raw retrieval scores stay similar [P3]. In contrast, the Aligned RAG Evaluation framework shows that standard attack-success-rate metrics mask both prevented and newly introduced failures in poisoning defenses, revealing that defenses like TrustRAG may prevent many attacks at low poison rates but also cause equal numbers of false positives, while detection-based defenses degrade sharply as poison density rises [P4]. These studies converge on the need for more nuanced evaluation beyond simple accuracy or ASR numbers.

Evaluation methodology itself is a recurring theme. Beyond poisoning, P5 introduces statistical tests (POB-ICM and POB-UNBP) to detect positional bias in LLM-based recommender RAG pipelines, demonstrating that bias persists across model sizes and cannot be eliminated by temperature tuning. Together with the Aligned RAG Evaluation benchmark, these works expose hidden failure modes in RAG systems that standard QA or retrieval metrics overlook, urging the community to adopt richer, multi-dimensional benchmarks.

Overall, the set reveals a split between engineering advances that enrich retrieval representations (P1, P2) and analytical advances that critique how we measure grounding, faithfulness, and bias (P3, P4, P5). The former achieve concrete performance lifts on specific multimodal tasks, while the latter call for broader, more diagnostic evaluation frameworks across modalities.

4 theme sentence(s) checked against the paper summaries.

At a glance

[P1] Pix2Evidence: A Multimodal RAG System for Text and Visual Evidence · relevance 0.46
Pix2Evidence is a multimodal retrieval-augmented generation system that fuses screenshot images with their transcriptions to improve answer accuracy and evidence selection on Wikipedia QA tasks.

[P2] Hierarchical Dual-Graph Retrieval for Long-form Audio Reasoning · relevance 0.42
AudioHelix introduces a hierarchical DualGraph retrieval-augmented generation framework that

consistently outperforms full-audio processing and prior RAG methods on long-form audio reasoning benchmarks.

[P3] Context as Infrastructure: Knowledge Compilation for Retrieval in Agentic Systems with Harnesses · relevance 0.42
The paper argues that retrieval-augmented generation needs a "knowledge compilation" layer to supply disciplined, compact context, and shows that a claim-evidence navigation tree can slash token use by up to 91 % and cut latency from 140 s to 30 s, though structural artifacts give no accuracy boost and the findings are limited to a single benchmark and model.

[P4] Prevented is Not Protected: What Attack Success Rate Hides about RAG · relevance 0.42
The paper shows that evaluating RAG poisoning defenses solely by attack success rate hides both prevented attacks and new failures, and proposes an "Aligned RAG Evaluation" framework that reveals these hidden effects.

[P5] Statistical Detection of Positional Bias in LLM-Based Recommender Systems: Methodologies and Empirical Evidence · relevance 0.34
The paper reveals that top-1, single-turn LLM-based recommender systems systematically favor items based on their prompt position, and introduces two statistical tests that reliably detect and quantify this positional bias across multiple state-of-the-art models.

[P1] Pix2Evidence: A Multimodal RAG System for Text and Visual Evidence

Ziniu Liu, Xiaoyang Chen, Dezhi Ye, Jie Liu et al. · arXiv 2610.41335v1 · https://arxiv.org/abs/2610.41335v1 · relevance 0.46

TL;DR: Pix2Evidence is a multimodal retrieval-augmented generation system that fuses screenshot images with their transcriptions to improve answer accuracy and evidence selection on Wikipedia QA tasks.

Problem
Retrieval-augmented generation over visually rich documents must locate both relevant content and the supporting evidence, but screenshot-based retrieval preserves layout while written facts remain hard to search, and topical relevance alone does not guarantee evidence relevance[1].

Method
Pix2Evidence encodes each screenshot tile together with its available transcription into a single fused image-text representation using a shared encoder f?, enabling joint consideration of layout and text during retrieval[3]. A FAISS index stores the normalized vectors for efficient similarity search[3]. A vision-language teacher model generates descriptions of answer-supporting evidence, which are used to adapt the retriever and to rerank retrieved candidates before answer generation[1][6].

Key results
- Without reranking, Pix2Evidence outperforms text-extraction and screenshot-retrieval baselines on four of five Wikipedia QA benchmarks and achieves 26.0 % accuracy on MMSearch and 40.6 % on EVQA, surpassing PixelRAG by 4.4 percentage points on MMSearch[4].
- With the vision-language reranker, answer accuracy improves to 33.2 % on NQ, 27.1 % on NQ-Tables, and 77.6 % on SimpleQA, representing gains of 2.1, 1.3, and 4.8 percentage points over PixelRAG respectively[4].
- The reranker also raises SimpleQA evidence recall from 83.1 % to 86.9 % and NQ-Tables recall from 50.7 % to 55.8 %[4].
- Modality ablation shows the best SimpleQA score (73.4 %) when both image and text are used in the retriever and reader, outperforming text-only (71.9 %) and image-only (67.9 %) configurations[4].
- Reading-depth experiments indicate that at k = 3 the non-reranked system reaches 26.0 % on MMSearch versus 21.6 % for PixelRAG, a 4.4-point gain[5].
- In the MoNaCo agentic search setting, the dual-view search on Pix2Evidence's fused index without

reranking attains the highest judge-F1 of 0.4816, improving over the base index by 0.0108[8].

Limitations
The evaluation is limited to English Wikipedia QA and a small set of model configurations, excluding other architectures, larger scales, and non-English domains[7]. The fused image-text index and vision-language reranker add storage and computation overhead, and the study does not measure peak memory usage or end-to-end latency, leaving a full cost analysis for future work[7].

Sources

[1] arXiv 2610.41335v1.pdf, arXiv abstract

[3] arXiv 2610.41335v1.pdf, page 4

[4] arXiv 2610.41335v1.pdf, page 6

[5] arXiv 2610.41335v1.pdf, page 7

[6] arXiv 2610.41335v1.pdf, page 8

[7] arXiv 2610.41335v1.pdf, page 9

[8] arXiv 2610.41335v1.pdf, page 17

[P2] Hierarchical Dual-Graph Retrieval for Long-form Audio Reasoning

Eunsang Lee, Taein Ju, Jiyoung Lee · arXiv 2610.41390v1 · https://arxiv.org/abs/2610.41390v1 · relevance 0.42

TL;DR: AudioHelix introduces a hierarchical DualGraph retrieval-augmented generation framework that consistently outperforms full-audio processing and prior RAG methods on long-form audio reasoning benchmarks.

Problem
Large audio language models (LALMs) struggle with long-form audio reasoning because processing entire recordings limits their ability to capture query-relevant content that is distributed across distant segments and incurs high computational and memory costs [1][2].

Method
AudioHelix builds a multimodal DualGraph offline for each recording, segmenting audio into fixed-length chunks and further into acoustic units, then extracting textual descriptions and acoustic embeddings to create three hierarchical levels: chunk-, unit-, and semantic concept-level nodes [2][3][4]. Edges are assigned weights as described, relevance is propagated from query-matched concept nodes through the graph using Personalized PageRank, and then the top-K audio chunks and acoustic units are retrieved for LALM generation [5].

Key results
On three long-audio benchmarks, AudioHelix achieves the highest or second-highest scores across three LALMs:

- **Audio-Flamingo-3**: 59.4 (TraceAV-Bench), 52.3 (MMAU-Pro), 58.5 (AudioMarathon) vs. LinearRAG 56.9/50.7/57.3 and PlanRAG-Audio 37.5/34.0/30.8 [5].
- **Voxtral-3B**: 50.6, 56.5, 57.9 vs. LinearRAG 48.1/40.9/56.4 and PlanRAG-Audio 44.6/35.4/39.0 [5].
- **Qwen3-Omni-30B**: 71.6, 67.2, 66.1 vs. LinearRAG 71.3/66.1/63.7 and PlanRAG-Audio 55.3/34.4/37.0 [5].

These results show that AudioHelix consistently improves accuracy (TraceAV-Bench, MMAU-Pro) and F1 score (AudioMarathon) while using only a small subset of the original audio [6].

Limitations
The provided excerpts do not discuss any limitations of the approach.

Sources

[1] arXiv 2610.41390v1.pdf, arXiv abstract

[2] arXiv 2610.41390v1.pdf, page 1

[3] arXiv 2610.41390v1.pdf, page 2

[4] arXiv 2610.41390v1.pdf, page 2

[5] arXiv 2610.41390v1.pdf, page 3

[6] arXiv 2610.41390v1.pdf, page 3

[P3] Context as Infrastructure: Knowledge Compilation for Retrieval in Agentic Systems with Harnesses

Zhichang Yu, Baining Cao, Sbaaoui Idriss, Qi Wang et al. · arXiv 2610.41172v1 · https://arxiv.org/abs/2610.41172v1 · relevance 0.42

TL;DR: The paper argues that retrieval-augmented generation needs a "knowledge compilation" layer to supply disciplined, compact context, and shows that a claim-evidence navigation tree can slash token use by up to 91 % and cut latency from 140 s to 30 s, though structural artifacts give no accuracy boost and the findings are limited to a single benchmark and model.

Problem
Retrieval-augmented generation is becoming agentic, but naïve search loops repeatedly issue queries, reread documents, and fill the model's context window with unused passages, leading to token waste and degraded answer quality [1][2].
The authors identify a missing "knowledge compilation" layer that would enrich source documents with structured context (e.g., claim-evidence pairs, entity graphs, timelines) before query time, separating the supply side from the harness that controls retrieval [1][2][3].

Method
The paper introduces a configurable ingestion pipeline that builds typed artifacts--most notably a navigation tree whose leaves are atomic claim-evidence pairs--and a retrieval harness (RAGFlow) that decides which artifact to feed to the model and when to stop the loop [1][2][3].
The system is open-source (https://github.com/infiniflow/ragflow) and operates on English corpora using BM25/grep-based harnesses and the MiniMax M3 model [1].

Key results
With only BM25/grep and MiniMax M3, the harness answers 85 % of BrowseComp-Plus questions and 91 % of multi-hop FRAMES questions [1].
A claim-evidence navigation tree reduces total query tokens by 91 % when the pre-fetched evidence is sufficient, adds 1.9 % when it is insufficient, and yields an average token reduction of 38 % [1].
The break-even sufficiency rate is 2 %, while the measured sufficiency on FRAMES is about 43 % [1].
A decision-gate mechanism lowers end-to-end latency from 140 s to 30 s [1].
Structural artifacts such as RAPTOR-style summary trees, PageIndex outlines, LLM wikis, and entity-relation graphs produce no detectable accuracy gain on FRAMES, and a PageIndex outline can become detrimental at scale [1][2].

Limitations
All measurements come from a single benchmark (FRAMES) and one compiler model, with no confidence intervals or repeated seeds reported [6][7].
The study's sample size is small, so only large effects (e.g., ~50 % token reduction, ~3× latency increase) are reliably observed [7].
Results are confined to English text corpora, BM25/grep harnesses, and the MiniMax M3 model; other

languages or retrieval methods are not evaluated [2].
The deployment ledger shows that none of the artifacts have yet met pre-registered targets in production, so no deployment-time benefits are demonstrated [5].

Sources

[1] arXiv 2610.41172v1.pdf, arXiv abstract

[2] arXiv 2610.41172v1.pdf, page 2

[3] arXiv 2610.41172v1.pdf, page 4

[5] arXiv 2610.41172v1.pdf, page 8

[6] arXiv 2610.41172v1.pdf, page 11

[7] arXiv 2610.41172v1.pdf, page 11

[P4] Prevented is Not Protected: What Attack Success Rate Hides about RAG

Chahana Dahal, Zuobin Xiong · arXiv 2610.40742v1 · https://arxiv.org/abs/2610.40742v1 · relevance 0.42

TL;DR: The paper shows that evaluating RAG poisoning defenses solely by attack success rate hides both prevented attacks and new failures, and proposes an "Aligned RAG Evaluation" framework that reveals these hidden effects.

Problem
Retrieval-augmented generation (RAG) systems can be compromised by knowledge-poisoning attacks that manipulate retrieved evidence to steer model outputs, and current defenses are judged mainly by attack success rate (ASR), which obscures defense-induced failures and the true mechanisms of "prevention" [1].

Method
The authors introduce Aligned RAG Evaluation, a matched-query protocol that jointly examines clean and poisoned responses with and without a defense, using matched-conditions analysis, LLM-judge endorsement, and retrieval-dependence controls to separate prevented from introduced failures [1, 6].

Key results
- On HotpotQA with one injected poison passage (p = 1), the TrustRAG adapter prevents 20 target-positive answers but also introduces 20, leaving the final target-positive count unchanged at 60/100; KeywordAgg shows the same 15-prevented/15-introduced pattern [4].
- With five poison passages (p = 5), TrustRAG prevents 65 targets and introduces only 1, reducing the target-positive rate from 96/100 to 32/100; similar defensive behavior is observed on Natural Questions (NQ) [4].
- Detection-based defenses (score-gap and score-dispersion detectors) intercept 59 of 60 target-positive outputs on HotpotQA at p = 1 (1 % false-positive rate) but intercept none of the 96 target-positive outputs at p = 5; on NQ they intercept 32/64 (or 36/64) at p = 1 but only 1/92 at p = 5 [5].
- Human audit of 100 outputs shows the LLM judge agrees with human annotators on 64 of 95 decisive cases (~ 68 % agreement) and 69 of 100 when including 5 "not applicable" cases; overall agreement drops to 50/98 for answer correctness when uncertain labels are counted as disagreements [5].

Limitations
The experimental PoisonedRAG cells contain only 100 queries each with injected passages shared in groups of ten, requiring group-level bootstrap intervals rather than treating queries as independent [7]; the evaluated defenses are adapted versions of KeywordAgg, TrustRAG, and prompt-control pipelines, not exact reproductions of the original papers [7]; and high uncertainty rates in clean-answer correctness (e.g., 147/200 uncertain TrustRAG responses on HotpotQA) prevent reliable conclusions about utility on benign queries [5].

Sources

[1] arXiv 2610.40742v1.pdf, arXiv abstract

[4] arXiv 2610.40742v1.pdf, page 5

[5] arXiv 2610.40742v1.pdf, page 7

[6] arXiv 2610.40742v1.pdf, page 8

[7] arXiv 2610.40742v1.pdf, page 9

[P5] Statistical Detection of Positional Bias in LLM-Based Recommender Systems: Methodologies and Empirical Evidence

Carlos Peláez-González, Francisco Herrera, Andrés Herrera-Poyatos · arXiv 2610.37962v1 · https://arxiv.org/abs/2610.37962v1 · relevance 0.34

TL;DR: The paper reveals that top-1, single-turn LLM-based recommender systems systematically favor items based on their prompt position, and introduces two statistical tests that reliably detect and quantify this positional bias across multiple state-of-the-art models.

Problem
Large Language Models are increasingly used as generators in conversational recommender systems, but exposing candidate items as an ordered list in the prompt can create structural positional bias, i.e., a systematic dependence of the recommended item on its position in the prompt [1]. This bias may cause less relevant items to be selected simply because of ordering, undermining recommendation quality.

Method
The authors propose two complementary detection methodologies: (i) POB-ICM, which uses contingency-matrix analysis and Chi-square independence testing, and (ii) POB-UNBP, a sample-efficient approach based on a uniformity null model with formal statistical guarantees [1, 2]. They implement a controlled retrieval-augmented generation (RAG) testbed that fixes all components except item order, using a real product catalog and a fixed prompt template [1, 4]. For each of 2 000 generated user queries they create 20 random permutations of the top-5 retrieved products (K = 5), yielding 40 000 query-permutation instances [1, 4]. Three metrics are introduced to quantify bias: permutation sensitivity (PS), positional bias (PB), and strong positional bias (sPB), the latter measuring recommendations that fall outside a query-dependent relevance set [1, 2, 4].

Key results
Both POB-ICM and POB-UNBP reject the null hypothesis of positional independence and uniformity for all evaluated models, confirming measurable bias [1]. Strong positional bias (sPB) across models ranges from 0.9 % to 10.2 % depending on model size and configuration; for example, DeepSeek R1 32B shows sPB = 2.9 % (95 % CI 2.8-3.1 %) while LLaMA 3.1 70B reaches sPB = 3.2 % (3.0-3.3 %) [7]. The lower bound L of the total variation distance to uniformity (TV) stays between 0.8 and 4.7 percentage points under stochastic decoding and between 1.0 and 4.9 points when both decoding noise and permutations are present, indicating that bias is not eliminated by temperature settings [7]. Paired Wilcoxon tests show that adding permutations to stochastic decoding significantly increases sPB for six of eight models by 0.3-1.1 percentage points (Holm-adjusted $p < 0.05$) [7]. Robustness analyses confirm that bias persists across different candidate-set sizes and decoding temperatures, suggesting a structural, model-dependent effect rather than an artifact of deterministic decoding [1]. The POB-UNBP test maintains high power: it rejects the null hypothesis on all configurations, with the largest p-value observed being 0.002 (DeepSeek R1 32B, top-5) and the lower bound L remaining stable (changes < 0.01 for most settings) [8].

Limitations
The study is limited to single-turn, top-1 recommendation scenarios in a controlled RAG testbed and does not evaluate multi-turn interactions or real-world deployment effects [1]. It focuses on detection rather than mitigation; mitigation techniques are identified as future work [6]. The experiments use product recommendation data with a fixed candidate-set size of five [4].

Sources

[1] arXiv 2610.37962v1.pdf, arXiv abstract

[2] arXiv 2610.37962v1.pdf, page 5

[4] arXiv 2610.37962v1.pdf, page 13

[6] arXiv 2610.37962v1.pdf, page 23

[7] arXiv 2610.37962v1.pdf, page 28

[8] arXiv 2610.37962v1.pdf, page 33